

Impacts of cacao agroforestry systems on climate change, soil conservation, and water resources: a review

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ABSTRACT

Agroforestry is crucial for improving water quality deteriorated by anthropogenic activities due to the use of chemical substances, including nitrogen (N) and phosphorous (P) in modern agricultural practices (MAPs). This state-of-the-art review aims to revise the influence of agroforestry systems (AFS) on climate change, soil conservation, and groundwater quality. The novelty of this review is that AFS can improve water quality, reduce nutrient loss, and support the biological, chemical, and physical properties of the soil. It is worth noting that the surficial geologic controls, slope gradient, soil types, and topographical conditions are factors that alter a watershed dominated by agroforestry areas. In addition, anthropic aspects, including agricultural practices, can also cause loss of water quality in basins dominated by an AFS area. This review is also novel in that it outlines how AFS can be used for the phytoremediation of contaminated soils to reduce aluminum (Al), mercury (Hg), and cadmium (Cd), among others. Therefore, AFS can be used for water decontamination, climate change mitigation, climate adaptation, and soil conservation. Further research is required to investigate the contribution of AFS to soil integrity.

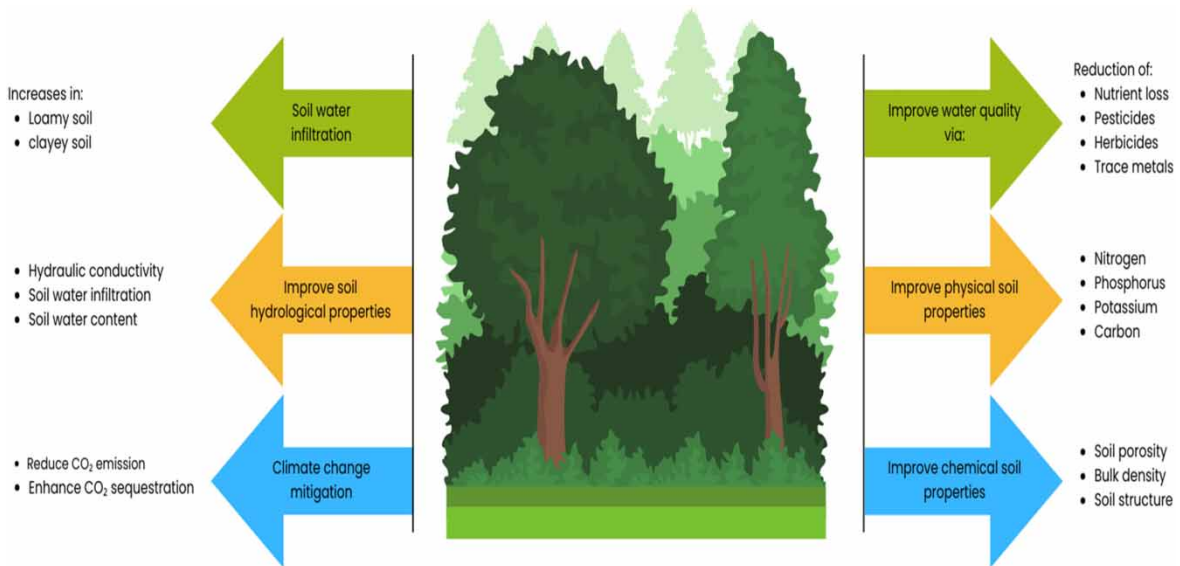
Key words: Agroforestry systems, Climate change, Ecosystem services, Soil conservation, Water quality

HIGHLIGHTS

- The role of agroforestry in water quality, soil conservation, and climate change mitigation was evaluated.
- The effectiveness of agroforestry systems (AFS) depends on the type of vegetation cover and soil properties.
- AFS is an effective technique to be used as pollution abatement for soil and water decontamination.

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GRAPHICAL ABSTRACT



1. INTRODUCTION

Agroforestry is practiced around the world and can deliver several environmental benefits (for instance, climate change mitigation and climate adaptation), soil conservation (e.g., erosion control, enhanced soil fertility) and ways of dealing with the problems posed by MAPs, such as loss of water quality caused by the overuse of chemical fertilizers (e.g., N and P) and pesticides. MAPs are a type of agricultural practice that are based on maximizing grain yields without respect to quality, the environment, or soil conservation. Fertilizers are broadly used in MAPs to improve the soil fertility (Kadigi *et al.*, 2020). Agroforestry is one of the most effective methods used to reduce a range of forms of abiotic stress, including drought, metal, and salinity resulting from a combination of climate change and the overuse of pesticides and chemical fertilizers, which have negative impacts on plant yields (Kumari *et al.*, 2022). Pesticides in forests and pastures contribute to soil degradation and are widely used by citizens, agroalimentary enterprises and cooperatives. In 2019, the global consumption of pesticides was estimated at 4.2 million tons (Foong *et al.*, 2022). Consequently, the severe impact of pesticide usage in agricultural activities has contributed to more than 50% and approximately 70% of freshwater eco-toxicity and human toxicity, respectively (Foong *et al.*, 2022). The chemical elements in soils cause eutrophication and thus the death of some aquatic organisms (Wear & Greis, 2002), as well as compromising the health and life of human beings. It was estimated that 1 out of 8, 1 out of 6, and 1 out of 4 people are at high risk from biochemical oxygen demand, N, and P pollution, respectively (International Food Policy Research Institute & Veolia, 2015). The Institute for Health Metrics and Evaluation (IHME) reported that 3.4% (80 million disability-adjusted life years (DALYs)) of the world's diseases are associated with inadequate water supply and sanitation (Weidema & Fantke, 2018). Drinking contaminated water is responsible for several diseases, including diarrhea, cholera, and hepatitis A. The Global Burden of Disease (GBD) estimated that 1,800,000 people died from diseases associated with water pollution in 2015 (GBD, 2015). Drinking contaminated water causes a total of 485,000 deaths each year as a result of diarrheal diseases (WHO, 2017). Researchers' attention has come to focus on effective

ways to deal with these challenges, taking into account the different roles of agroforestry associated with ecosystem services (ES), including groundwater quality.

Different to MAPs, agroforestry is a sustainable alternative that protects soils against erosion and enhances water quality (Neto *et al.*, 2007). However, there is uncertainty about the impact of cacao (*Theobroma cacao* L.) trees on water quality. Cacao is generally grown in the shade of a thinned forest (Lobão *et al.*, 2007) and is mainly grown in tropical areas of Central and South America, Asia, and Africa (Marita *et al.*, 2001). Cacao is one of the most important perennial crops in the world. Cacao production worldwide was estimated at approximately 3.5 million tons in 2006 (ICCO, 2007). In 1970, the production of cacao represented approximately 0.6% of Brazil's Gross National Product (Fonseca *et al.*, 2020). In 2015, the production of cacao in the southeastern region of the state of Bahia, Brazil, represented 87% of the agricultural land (Sanches, 2019). Integrated cacao agroforestry is a sustainable solution, combining forests and agricultural activities to reduce runoff and soil erosion, mitigate climate change through the storage of carbon, provide goods to human beings, and promote conservation and biodiversity. As compared to the preserved forest, CO₂ emissions in cacao-AFS are variable and depend on soil characteristics attributed to the type of vegetation cover (Costa *et al.*, 2018). Likewise, agri-silvi-based AFS had higher soil moisture than sole crop fields (Bhatt *et al.*, 2016). Other researchers have argued that AFS have a great influence on the supply of regulating ES and enhancing landscape structure (Kay *et al.*, 2018). The adoption of agroforestry can present benefits that significantly differ from other land use. For example, hedgerow-based AFS is more efficient than other land uses regarding its role in reducing surface runoff (Bhatt *et al.*, 2016). AFS soil quality does not differ from that of the forest (Rousseau *et al.*, 2012), and can therefore contribute to the regulation of the water cycle and modify soil water availability.

Forests can act as water filters depending on the forest extensions and forest types. For instance, riparian forests contribute to clean water owing to their positive influence on pesticide removal in water-saturated zones (Aguiar *et al.*, 2015). The researchers argued that wood > shrubs > grass in terms of forest performance for removal of pesticides in the riparian zones. As such, riparian vegetation has a fundamental role in the ES associated with water quality. In other words, agroforestry can play a crucial role in restoring riparian habitats and solving river problems. In addition, agroforestry can be seen as encouraging environmental protection and biodiversity conservation (Nair, 2011). For example, vegetative filter strips (VFS) can be planted in an area with the purpose of removing sediments, nutrients, and pesticides from both surface and subsurface waters (Dillaha *et al.*, 1989; Nair & Graetz, 2004; Nair, 2011). The VFS play an impressive role in controlling erosion rate and maintaining soils in the field (Grismer *et al.*, 2006). Also, AFS are suitable strategies for soil conservation and infiltration recovery. Soil water content (SWC) can be increased in the agroforestry areas depending on the type of tree, slow-growing or fast-growing trees, and soil composition. Taking into account the importance of AFS for environmental challenge mitigations, and their importance for ES, this state-of-the-art review investigated the influence of AFS on climate change, soil conservation, and groundwater quality. We hypothesize that: (i) agroforestry can control erosion and improve water quality; and (ii) agroforestry can be used for pollution abatement in soils and groundwater. This review proceeds as follows: the introduction, importance of AFS for soil conservation, the role of AFS in enhancing water infiltration, the importance of AFS on soil chemical properties, role of AFS in reducing climate change, influence of AFS on water quality, future challenges and ways forward, and finally conclusions.

2. IMPORTANCE OF AFS FOR SOIL CONSERVATION

Finding a sustainable solution to deal with soil degradation is becoming more and more crucial to maintaining the conservation and integrity of the soil. Soil integrity refers to the capacity of soil to perform its essential functions (Karlen *et al.*, 1997), whereas soil conservation is a way of controlling soil erosion and maintaining soil fertility

(Young, 1989), which is crucial for the environment. AFS has an effective role in improving physical properties (Figure 1) by adding organic matter, including soil structure, and porosity (Kumar *et al.*, 2020). It has been discovered that AFS improve soil structure through the plant material of tree components and the addition of organic material (Muñoz-Rodríguez *et al.*, 2020).

Research implemented in Pasto city, Colombia, has shown that the distribution of aggregates changes from 60 to 70.3% before and after agroforestry interventions (Muñoz-Rodríguez *et al.*, 2020). This shows that AFS contribute to strengthening soil structure, and avoid runoff transporting agricultural soil to the sea, lakes, or rivers (Figures 1 and 2). Noticeably, trees enhance the quantity of water used on-farm for trees and increase the biomass of trees or crops produced per unit of water (Ong, 2006). Another study reported that bulk density, porosity, field capacity, and wilting point varied over the years in agroforestry areas (Arévalo-Gardini *et al.*, 2015). Along the same line, research on pecan AFS has shown that soil bulk density decreased from 16.13 to 7.10%; however, soil content and total porosity increased (Wang *et al.*, 2022). Soil bulk density can vary from one soil type to another, depending on the duration of agroforestry. It is worth noting that soil bulk density, noncapillary porosity,

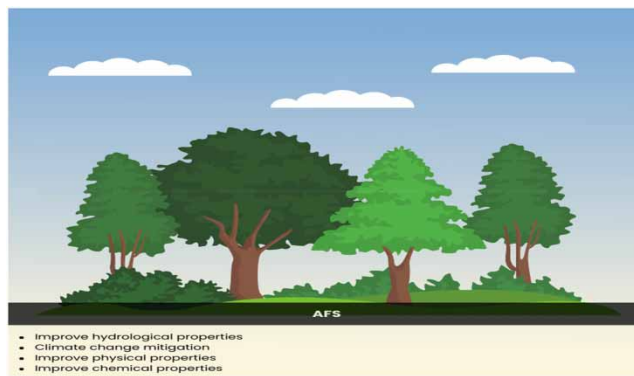


Fig. 1 | Benefits of AFS regarding soil conservation and climate change mitigation.

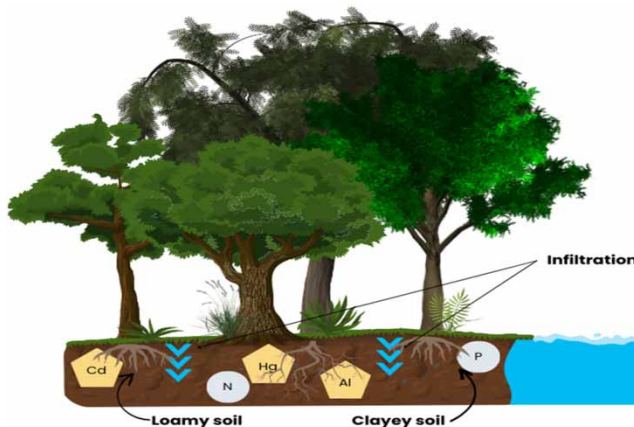


Fig. 2 | Role of AFS to enhance soil water infiltration.

and capillary porosity are often influenced by a set of factors, such as field capacity, saturated water holding capacity, and saturated hydraulic conductivity (Jiang *et al.*, 2019). AFS are crucially in rainy regions, and the mountains to slow down heavy rainfall and infiltrate it into the soil, and prevent farmers from losing their crop yields, respectively.

2.1. Importance of AFS for ecosystem services

Among numerous alternatives that can be used for natural resources management, agroforestry is one of the most sustainable and environmentally friendly. Agroforestry enhances ES via improved soil structure and increases water retention and carbon sequestration (Mukhlis *et al.*, 2022), which can be possible thanks to the decomposition of fine roots of trees (Siegwart *et al.*, 2022). As such, the carbon sequestration rate might depend on soil depth. For example, topsoils, upper soil, and lower subsoils can potentially influence carbon sequestration in AFS in the following order: topsoils ($0.92 \text{ Mg ha}^{-1} \text{ yr}^{-1}$) > upper subsoils ($0.72 \text{ Mg ha}^{-1} \text{ yr}^{-1}$) > lower subsoils ($0.52 \text{ Mg ha}^{-1} \text{ yr}^{-1}$) (Hübner *et al.*, 2021). Moreover, research carried out in the West African Sahel has concluded that agroforestry can promote landscape resilience in the greater region (Ellison & Ifejika Speranza, 2020). For example, AFS have reduced soil loss from 1.22 to 0.17 t/ha over a 6-year period (Panwar *et al.*, 2018). This reduction might be possible due to the capacity of cacao trees to maintain and improve soil structure. These findings explain that agroforestry has great importance for soil conservation. As such, AFS promote sustainable development (i.e., functional ecosystems, livelihoods, and human security) through the accomplishment of goals including providing water, energy, and food security, among others (Elagib & Al-Saidi, 2020). Importantly, AFS also protect land use and maintain good physical, chemical, and biological properties in tropical soils (Arévalo-Gardini *et al.*, 2015, 2020; Matos *et al.*, 2020), and contribute to the restoration of a local landscape scale (Martin *et al.*, 2021). However, these properties can be improved if cover crops and trees are included in the system (Alegre & Cassel, 1996).

Similarly, scholars underscored that the biological, chemical, and physical characteristics of forest soils contributed to high quality water in streams, and moderated stream hydrology (Neary *et al.*, 2009). Research conducted in Ejido, Mexico, has demonstrated that cacao agroforests have lower land change as compared to other types of land tenure (Oporto-Peregrino *et al.*, 2020). Cacao agroforestry, it is worth noting, is characterized by a large accumulation of the amount of soil organic carbon (SOC) (Monroe *et al.*, 2016), and it can be used to reduce deforestation and forest degradation (Batsi *et al.*, 2020), improve soil fertility, maintain soil ecological functions, and restore soil quality to degraded pasturelands (Suárez *et al.*, 2021).

3. IMPORTANCE OF AFS FOR SOIL CHEMICAL PROPERTIES

In recent years, studies have been carried out to search for sustainable routes to maintain the soil's physicochemical properties, which are essential for its conservation and productivity. Research conducted in Nepal showed that there were significant differences in soil pH, Al content, base saturation, electrical conductivity, organic matter and N content, and cation exchange capacity between AFS and conventional system (CS) (Schwab *et al.*, 2015). This finding was consistent with the finding of research conducted by Tornquist *et al.* (1999) in Costa Rica, which concluded that soil exchangeable bases and pH were lower in agroforestry treatments as compared with pastures. This indicates that AFS have higher soil quality and are more fertile than CS. AFS have more ability to ensure plant and animal health, and to preserve environmental quality, as compared to CS. This finding is supported by another study carried out in the central mid-hills of Nepal by Schwab *et al.* (2015). AFS reduce surface runoff, SOC, and associated nutrient losses by an average of 58, 65, 9, and 50%, respectively (Zhu *et al.*, 2020). The reduction of runoff by the AFS is a part of soil conservation and therefore maintains the chemical and physical properties of soil (Figure 1). Tree roots in AFS reduce N and P residues in soils by 20–100% (Pavlidis &

Tsihrintzis, 2018). Findings of other research have also reported that AFS can reduce pesticide leaching and runoff by up to 90% (Pavlidis & Tsihrintzis, 2018). This is possible due to soil and water management in this system (Wang *et al.*, 2017), which can potentially reduce non-essential trace metals in soil. For example, cacao trees enable the reduction of non-essential trace metals such as Cd and thus transfer it from roots to the shoot (Oliveira *et al.*, 2022) (Figure 2). Another study conducted in the province of Bagua in Amazonas, Peru, indicated that the amount of Cd concentrated in the roots of cacao trees is five times higher than the Cd level in the soils (Oliva *et al.*, 2020). In the same line of reasoning, another study conducted in Brazil highlighted that AFS improved soil chemical indicators based on pH increase, soil nutrient content (Ca, Mg, and K), reduced Al saturation (Silva *et al.*, 2020), and Hg (Béliveau *et al.*, 2017) (Figures 1 and 2). Notably, AFS contribute to the maintenance of soil integrity (Béliveau *et al.*, 2017). These findings indicate that cacao-AFS is effective at reducing trace metals in soils that are often caused by MAPs. Accordingly, AFS are an effective technique to be used as pollution abatement for soil and water decontamination (Figure 1). For instance, trees or woody plants with developed root systems and large biomasses are attractive for vegetation and phytoremediation in metal-polluted sites (Lee *et al.*, 2009; Capuana, 2013).

4. ROLE OF AFS IN ENHANCING WATER INFILTRATION

AFS play a crucial role in reducing the need to drain, soil evaporation, runoff, erosion, and silting up of rivers (Figure 2). The reduction in runoff can be attributed to ground cover plants in AFS, and the slope of the landscape. For example, research carried out in the Philippines reported that the risk of soil erosion on slopes was 8% higher with non-agroforestry use than with agroforestry use (Delgado & Canters, 2012). Other researchers underlined that AFS increased aggregate soil stability, and decreased runoff and soil erosion (Roose & Ndayizi-giye, 1997; Marwah, 2012), due to enhanced infiltration (Mwangi *et al.*, 2016), which resulted in increasing soil water storage (Zhao *et al.*, 2022). Scholars have pointed out that agroforestry can increase evapotranspiration (ET) in the watershed (Wang *et al.*, 2017) and thus reduce water availability owing to water uptake by tree roots (Mwangi *et al.*, 2016), which is strongly related to root-length densities and root surface areas (Bayala & Prieto, 2020). Similarly, findings of another study highlight that apple fine roots decrease the SWC (Shen *et al.*, 2022). Importantly, the ability of trees to reduce the SWC depends on tree and soil types. For example, a previous study demonstrated that the SWC decreased rapidly during the summer in Putnam silt loam dominated by agroforestry buffers (Sahin *et al.*, 2016). In contrast, the walnut-wheat alley cropping system (JTACS) increased water infiltration in the shallow soil layer during the rainfall season (Wang *et al.*, 2015). Water availability through soils and air moisture is one of the main drivers of AFS (Pérez-Girón, 2022). In the case of cacao trees, the density of lateral roots exponentially decreases with depth, and 20% of these roots can uptake water (Mommer, 1999). Research carried out in a coastal area of northern Iran has indicated that the combined scenarios of agroforestry, no-tillage, and rice straw mulch are an effective way to increase groundwater recharge in the region (Mohseni *et al.*, 2022). Thus, AFS has among its functions the regulation of rainwater, support of the microclimate (crop production), and ecosystem stability (Theresia Sri Budiastuti *et al.*, 2021).

4.1. Factors influencing soil water availability

The effectiveness of agroforestry to increase or reduce the SWC may depend on the type of vegetation cover, and soil properties. Vegetation density and type of vegetation are correlated with soil hydraulic properties (Reddy *et al.*, 2016). Similarly, organic matter produced by AFS enhances the effective rewetting and water retention capacity (Lestari & Mukhlis, 2021).

Agroforestry reduces excessive evaporation and maintains groundwater availability during the dry season (Lestari & Mukhlis, 2021). It also helps improve soil quality by cooperating with water stress and improving

hydraulic conductivity and infiltration, owing, perhaps, to grass and cover crops. Therefore, agroforestry is a strategy to be used to control runoff and erosion. Infiltration has a major impact on controlling soil erosion and runoff, soil moisture content, and groundwater recharge (Lozano-Baez *et al.*, 2019; Figure 2). Depending on the precipitation regime, runoff carries chemical substances from an agricultural field into the watershed (Figure 2). Consequently, syntropic agroforestry systems, which are a type of agroforestry involving very large amounts of organic matter, can be used to improve soil water infiltration (Murta *et al.*, 2021). In addition, there are other solutions such as covering the soil, rather than leaving the bare soil. Recent research has highlighted that AFS reduce the total soil loss and runoff by 37.7 and 19.1%, respectively (Jinger *et al.*, 2022).

4.2. Relationship between tree roots and water infiltration

There is a complex relationship between soil water infiltration and tree roots. For example, trees with deep roots increase macropores vertically and modify both micro and mesopores laterally (Wang *et al.*, 2017) as well as being more suitable to improve soil moisture content as they are able to lift up or redistribute water to the topsoil via a process known as hydraulic lift (Bayala & Prieto, 2020). Both hydraulic lifting processes and deep roots allow trees to uptake the groundwater from the deeper layer and make it available to understory vegetation with shallow roots (Pérez-Girón, 2022). In addition, deep-root trees improve soil physical conditions and higher soil microbiological activities under AFS (Nair *et al.*, 2008). Thus, the infiltration depends on the land use history prior to the application of agroforestry in an area (Lozano-Baez *et al.*, 2019). For instance, mechanical clearing reduces infiltration rates because the infiltration can depend on the woody vegetation roots. As such, the infiltration rate of soil increase is positively linked with the enhancement of soil mechanical stability (Schnug & Haneklaus, 2002). Furthermore, the greater the infiltration capacity of buffer soils, the more agroforestry species are able to remove pesticides from overland runoff (Dollinger *et al.*, 2019). Infiltration may vary with agroforestry practices and plants used and may be affected by plant age, which could reduce baseflow and water yield in the soil or watershed due to a possible increase in ET of some agroforestry species.

4.3. Relationship between water infiltration and soil characteristics

Soil moisture may depend on a range of factors, including the type of land cover and soil characteristics. The uppermost soil layer in the AFS has higher soil moisture as compared to monoculture (Niether *et al.*, 2017). On the other hand, Sarto *et al.* (2022) highlight that AFS reduce water availability in the superficial soil layer, particularly during the rainy seasons. These findings indicate that the capacity of trees to recharge the groundwater depends on several factors, including the climate, soil types (e.g., sand, silt, clay, etc.), and soil characteristics, including fracture depth, soil texture, and fractured rock. Water in the soil can be influenced by pedotransfer functions (Reichert *et al.*, 2009), which have a great influence on infiltration and water quality. For example, research reported that loamy sites could have a water availability greater than clay loam or sandy (Dodd & Lauenroth, 1997). Similarly, another study indicated that infiltration can be lower in sandy soils than in clayey soils (Lozano-Baez *et al.*, 2019). Therefore, these findings indicate that a high infiltration rate depends on soil characteristics, including high soil aggregation. The quality and quantity of coastal waters result from a complex interaction of anthropogenic activities with soil and climate (Daly *et al.*, 2018). In other words, coastal and groundwater quality depend on soil properties and land use. The soil in a constructed wetland is an optimal substrate and has great performance in the adsorption and passivation of pollutants, including N, P, and heavy metals in water (Cheng *et al.*, 2021). Another study has outlined that a decrease in soil pH and salinity accompanied by a decrease in Cl and Na concentrations is registered in an area irrigated with treated wastewater combined with the plantation of agroforestry species (Zouari *et al.*, 2019).

5. INFLUENCE OF AFS ON WATER QUALITY

Groundwater quality depends on a complex system (land use, anthropogenic activities, and vegetation cover) in which the physicochemical properties of the soil have a major influence (Figure 1). Chemical substances used in MAPs cause the contamination of water. This challenge draws researchers' attention to focus on AFS, which includes advantages such as the improvement of water quality, carbon sequestration, biodiversity conservation (Zinkhan & Mercer, 1996; Jose *et al.*, 2009; Tomer *et al.*, 2009; Kumar *et al.*, 2010; Holzmüller & Jose, 2012; Lovell *et al.*, 2018; Alagele *et al.*, 2019), nutrient cycling and water retention (Sollen-Norrin *et al.*, 2020), and reductions in the use of chemicals fertilizers (Figures 1–3). In addition, AFS reduce water loss and wind velocity, and thus limit wind erosion (Kumar *et al.*, 2020). In contrast to AFS, MAPs can potentially exacerbate soil loss, contaminate groundwater, and cause pollution from other sources (Nair & Graetz, 2004; Schultz *et al.*, 2004). Such practices endanger human health and harm aquatic life because of the deterioration of water quality and eutrophication, respectively. Watersheds may contain suspended solids that are affected by erosional factors in runoff (Rodríguez-Blanco *et al.*, 2019). Certain substances may remain in the waterbody for long periods of time, whereas others remain for only a short time, depending on the substance and the precipitation regime. AFS reduce the losses of herbicides, pesticides, and other pollutants by 55–100% (Zhu *et al.*, 2020). The reduction occurs through the extensive lateral root systems that scavenge soil nutrients and thus redistribute them beneath tree canopies (Rhoades, 1996).

A study conducted in Spain indicated an agroforestry catchment feeds a water reservoir with a sedimentation rate estimated at 3.5 tons/ha per year (Palazón *et al.*, 2016). This amount of sediments depends on agricultural

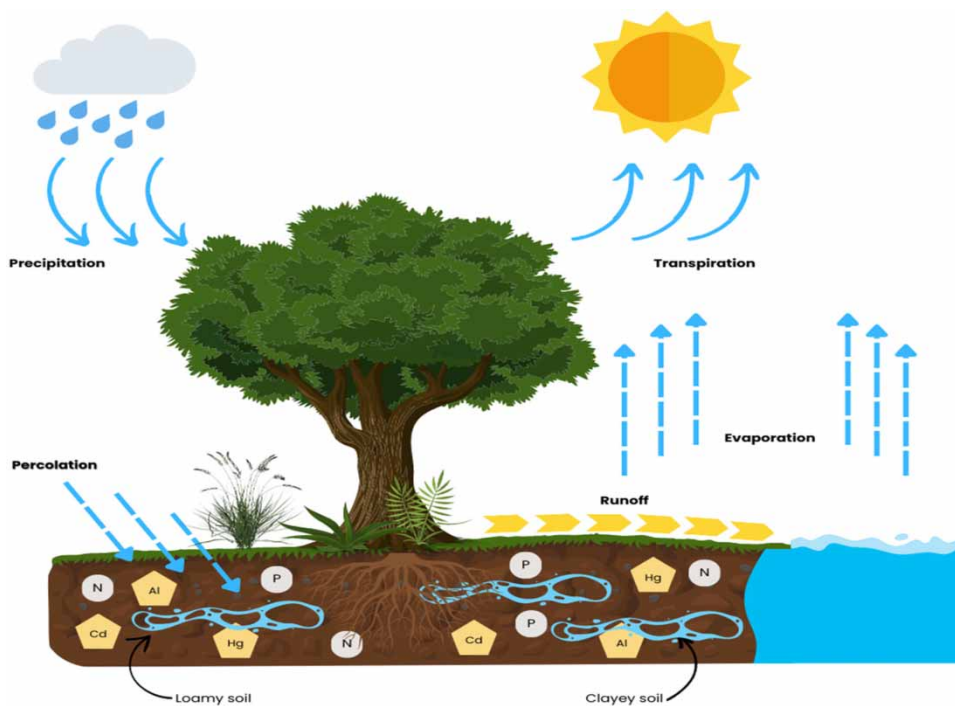


Fig. 3 | Role of AFS in improving water infiltration and reducing contaminants in soil.

practices, particularly the types of chemical substances used in the AFS. Pesticides can threaten groundwater quality if they are used in the root-affected zone (Dollinger *et al.*, 2019). Moreover, other factors (for instance, precipitation, slope, soil porosity, and buffer localization) can also play an important role in water pollution. According to Canencia *et al.* (2011), N, P, and potassium levels in watersheds dominated by AFS are moderately high and suitable for crop production. This demonstrates that agroforestry has great potential to remove these chemical elements.

5.1. Utilization of AFS as a phytoremediation technique

Watershed contamination is a worldwide problem that can become hypertrophic and thus cause the extinction of some species of aquatic organisms when these chemical elements are in high concentrations. These challenges have compelled researchers to search for facile and sustainable solutions, such as agroforestry. Kim & Cho (2014) found in their study that agroforestry crop fields and open crop fields had similar runoff loadings of total N, total P, and total suspended solids. Notably, AFS can reduce nonpoint source pollution due to tree growth and tree roots taking more space in the soil, reducing the amount of fertilizer in the watershed (Udawatta *et al.*, 2010) and generating cleaner water (Brown *et al.*, 2018) (Figure 3). As compared with other existing reviews, this study shows that, in an agroforestry area, trees with deep root systems have a great influence on the effectiveness of removing contaminants in soil and improving groundwater quality through trapping of nutrients, and metals deposited in the surface and topsoil (Kumar *et al.*, 2020; Pavlidis *et al.*, 2021). Therefore, agroforestry acts as a technique for the remediation of polluted soils owing to tree roots that can uptake the excess agrochemicals that would otherwise pollute groundwater via leaching and surface water via runoff (Pavlidis *et al.*, 2018, 2020, 2021). In other words, AFS can act as a technique to improve water quality, which is an objective of ES (Keeler *et al.*, 2012).

6. ROLE OF AFS IN CLIMATE CHANGE MITIGATION

The challenges of managing climate change are increasingly drawing researchers' attention to the need to find sustainable solutions. AFS can be an effective tool for mitigating climate change (De Zoysa & Inoue, 2014; Murthy *et al.*, 2016) and carbon and environmental footprints, by reducing greenhouse gas emissions into the atmosphere (Tefera *et al.*, 2019), regulating rainwater (soil and water conservation), and supporting the microclimate (Budiastuti *et al.*, 2021), due to the shade provided by this system (Lestari & Mukhlis, 2021). AFS are suitable for climate mitigation, particularly as they have a high density and diversity of shade trees (Boreux *et al.*, 2016). The adoption of AFS with 50% shade cover can reduce the mean temperature (Gomes *et al.*, 2020). AFS are climate-resilient agricultural practices (Jhariya *et al.*, 2019). Other researchers have pointed out that AFS can contribute to climate adaptation (Verchot *et al.*, 2007). Shade has an important role to play in the reduction of temperature and enhances rainfall (Panozzo *et al.*, 2022). The effect of trees on soil water conservation increases with the intensity of shading. Thus, the density and composition of the species have a great influence on climate mitigation. For example, coffee systems reduce the effect of extreme temperature and precipitation (Lin *et al.*, 2008), findings have shown that the shaded coffee AFS improve microclimate conditions and deep-water drainage as compared to unshaded coffee systems (de Carvalho *et al.*, 2021).

6.1. Contribution of AFS to natural resources management

People around the world are currently facing food and water insecurity owing to the depletion of natural resources, such as water and crops. This challenge has increasingly spurred scientists to seek out a simple and environmentally friendly solution. Scholars report that AFS maintain high levels of biodiversity and biomass (Sistla *et al.*, 2016), which are necessary for the integrity and conservation of the soil. Similarly, other researchers

argue that AFS combine production and conservation (Vallejo-Ramos *et al.*, 2016) and are the best option to deal with the erosion, and degradation of biodiversity (Saha *et al.*, 2010) usually caused by the MAPs. Scientists have reported that farmers are used to felling trees to enlarge their gardens and producing furniture (Pauleus & Aide, 2020; François *et al.*, 2022). Importantly, AFS can reduce deforestation and cultivable land extension (François *et al.*, 2022), protect environmental areas (Laudares *et al.*, 2017), recycle nutrients lost in MPAs (Izac & Sanchez, 2001) via their root systems (Raj, 2020), lift people out of poverty and stem the decline of agricultural productivity (Jama *et al.*, 2006). In the other words, AFS is a promising and sustainable option for natural resources management and this can contribute to human well-being and environmental protection.

6.2. Importance of cacao-AFS for water loss reduction

Cacao agroforests are described as being sensitive to climate change and water deficit (Lahive *et al.*, 2019), do not cause undesirable environmental change (Gama-Rodrigues *et al.*, 2021), and have a crucial role in drought-tolerant land use (Gateau-Rey *et al.*, 2018), particularly in tropical regions (Schwendenmann *et al.*, 2010). Wang *et al.* (2017) have underscored that AFS increase vertical preferential flow and retard the subsurface lateral flow and thus enhance water retention capacity. Another study highlighted that cacao is tolerant to shade, and the maximum photosynthetic rate occurs at an irradiance of around $400 \mu\text{mol m}^{-2} \text{s}^{-1}$ (Arévalo-Gardini *et al.*, 2021). Admittedly, unshaded cacao monocultures are vulnerable to climate change (Heming *et al.*, 2022). In contrast, shade trees reduce water loss from top soil via ET of the cocoa (Niether *et al.*, 2017). Therefore, canopy structure has a great influence on the ET in AFS (Wang *et al.*, 2021). The age of trees may have a great influence on the capacity of AFS to enhance the advantages and ecological services.

6.3. Importance of cacao-AFS for climate change mitigation

AFS have an evident ecological advantage over monoculture, particularly for young apple trees in the semi-arid region (Zhao *et al.*, 2022). As compared to older cacao trees, shade is more necessary for young cacao tree plantations (Tscharntke *et al.*, 2011). In addition, AFS can both contribute to capturing soil carbon and play an important role in mitigating atmospheric CO_2 (Verchot *et al.*, 2007; Rita *et al.*, 2011). The total carbon (aboveground and root biomass) stored in an AFS, including cacao and shade trees, was 2.5 times higher than in a monoculture (Niether *et al.*, 2020). Shade trees have great importance in cacao-AFS because of their role to influence radiation, wind regimes, nutrients, and hydrological cycling (Tiralla *et al.*, 2013). Notably, shade trees within AFS protect the understory cacao against extreme climate (Niether *et al.*, 2018). Cacao agroforestry can make an important contribution to mitigating climate change and contribute to the national economy of a country. AFS can also be a promising solution for improving low soil fertility in tropical zones (Rangel Mendoza & Silva Parra, 2020) and contribute to the achievement of Sustainable Development Goal 2, which aims to end global hunger.

7. FUTURE WORK AND WAY FORWARD

It is crucial to find a sustainable and environmentally friendly alternative to deal with the problems of soil loss, nutrient loss, soil contamination, climate change, and water contamination. These challenges are drawing scientists' attention to focus more on AFS. Herein, this present review revised the influence of AFS on soil conservation, climate change, and groundwater quality. This study revealed that AFS offer a sustainable approach for dealing with climate change, groundwater contamination, and nutrient loss, among other environmental challenges. The strengths of this study encompass a range of evidence about the relationship between AFS and water quality and how environmental and geologic factors, as well as human activities, can be associated with agroforestry to contaminate the waterbody. Insufficient evidence about how woody plants can influence the intercept and/or reduction of heavy metals in soil is a limitation of this review. Another limitation is that it does not

take into account the effects of fast-growing and low-growing trees on soil water availability. This lack of evidence could be filled via an experiment carried out using fast-growing and low-fruit trees planted in arid, semi-arid, and/or humid sites where the SWC is measured before and after the observations. Such an observation requires consideration of the soil type and the different phases of the trees' growth as important factors. Researchers report that fast-growing trees are not suitable for forestation in areas with medium precipitation and brackish groundwater (Sudmeyer & Simons, 2008), and their photosynthetic rates and stomatal conductance are higher than slow-growing trees (Liu *et al.*, 2021). Moreover, the interaction between shape of tree roots, the type soil and trees' physiology should be studied to enlarge knowledge concerning trees' behavior towards infiltration or soil water availability. In conclusion, further studies are required to identify other factors that may influence the ability of AFS to improve water quality and to examine how agroforestry age and agroforestry density can influence soil water infiltration.

8. CONCLUSIONS

There is an urgent need to find a sustainable option for reducing soil loss, recharging groundwater, and improving water quality altered by industrial activities and MAPs. This review investigated the effects of AFS on soil conservation, climate change, and groundwater quality. It was found that AFS can enable soil water infiltration and ET. We also found that agroforestry is a promising and sustainable alternative for improving water and soil quality and controlling erosion. This review showed that AFS improve soil structure through the vegetal material of tree components. Also, AFS support soil biological, chemical, and physical properties. Therefore, AFS are crucial for soil conservation and the enhancement of water quality. In addition, AFS are a promising alternative to be used to remove trace metals (e.g., Cd, Al, Hg) in contaminated sites. However, several factors, including surficial geologic controls, slope gradient, soil types, and topographical conditions can combine with the AFS to deteriorate water quality in the watersheds. These factors can be accentuated by land use/anthropic activities. The deep-root trees play an important role in reducing pollutants in the soil and enhancing humidity. Our hypotheses were verified because we have found that AFS control erosion and improve water quality, and can be used as a phytoremediation technique to remove trace metals in the soils. The novelties of this review, compared to existing ones, are that AFS can contribute to reducing non-essential trace metals in soils caused by MAPs and improve soil quality by increasing infiltration, soil organic matter, macropores, aggregate stability, and aeration by decreasing erosion, runoff, and soil bulk density, and reducing nutrient loss. Conclusively, the adoption of AFS can be a promising alternative to improve upstream water quality, soil conservation, agricultural sustainability, mitigate climate change and allow carbon sequestration, and reduce environmental footprints.

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AUTHOR CONTRIBUTIONS

M.F.: Conceptualization, methodology, design, material preparation, data curation, writing – original draft, writing – review & editing, validation. E.M.-N.: Supervision, writing – review & editing, validation. M.C.G.P.: Writing – review & editing, validation. A.L.S.: Writing – review & editing, validation.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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